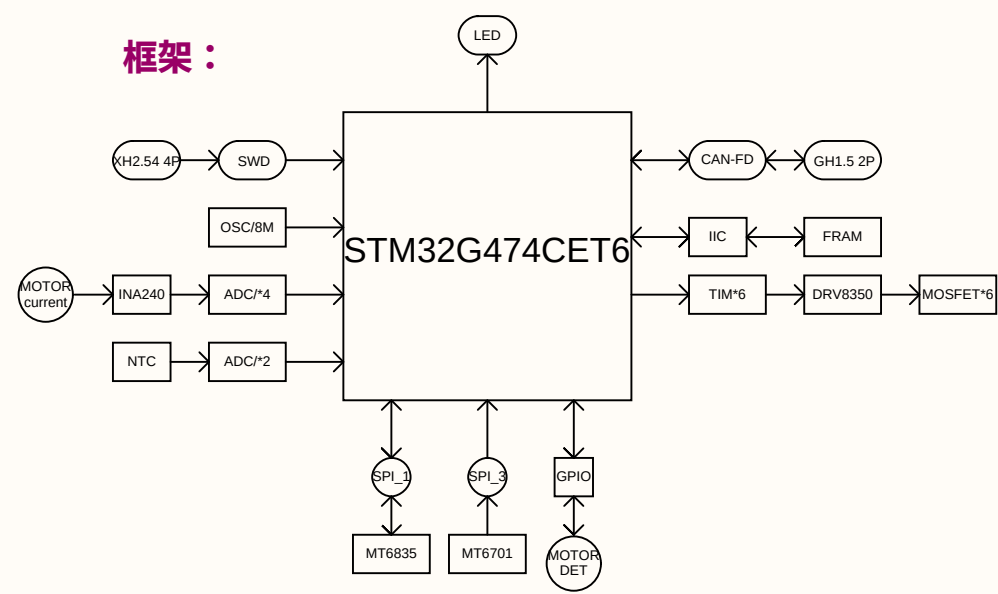
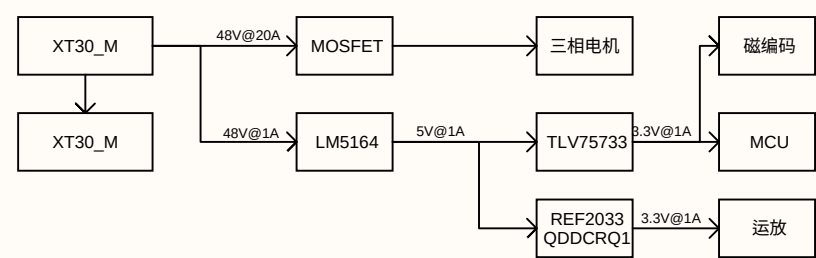


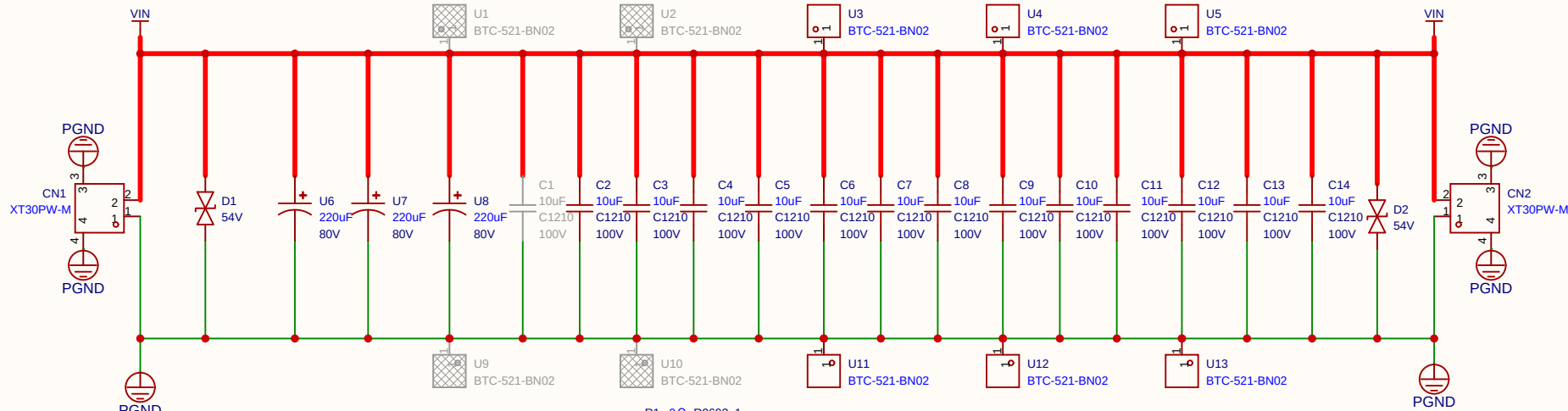
框架：



电源：



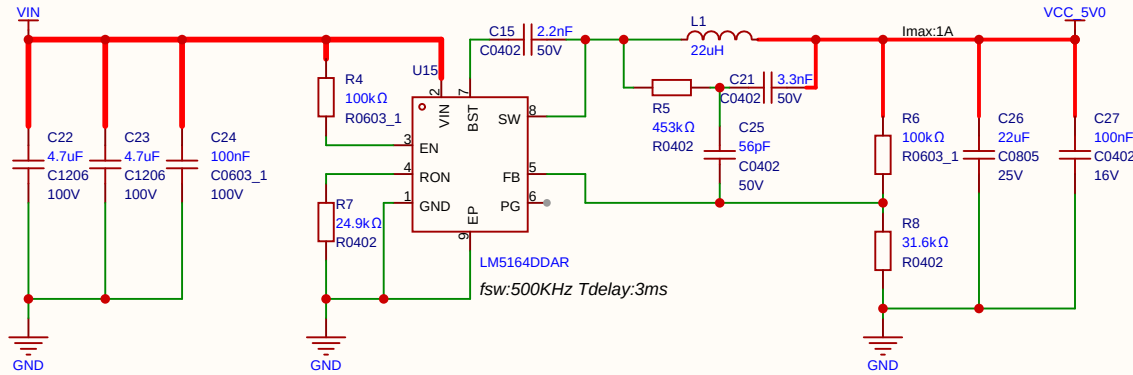
|                                                                                      |    |                  |           |       |            |     |
|--------------------------------------------------------------------------------------|----|------------------|-----------|-------|------------|-----|
| 原理图                                                                                  |    | FOC_SCH_V1.0     |           | 创建日期  | 2026-06-12 |     |
|                                                                                      |    |                  |           | 更新日期  | 2026-07-01 |     |
| 板子                                                                                   |    | FOC_PCBA018_V1.0 |           | 图页    | P1         |     |
| 绘制                                                                                   | FH |                  | 一体化关节_FOC |       |            |     |
| 审阅                                                                                   |    |                  |           |       |            |     |
|                                                                                      | FH |                  |           |       |            |     |
|                                                                                      |    | 版本               | 尺寸        | 页     | 1          | 共 8 |
|  |    | V1.0             | A4        | CHRIC |            |     |



电源输入滤波



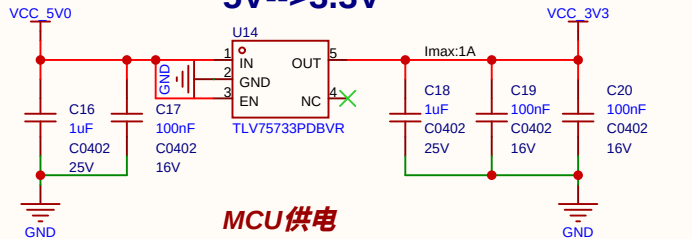
VIN-->5V



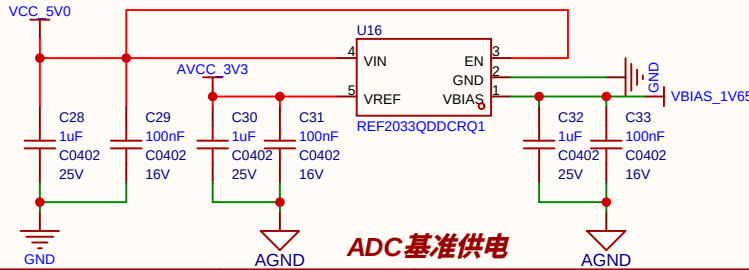
VFB=1.2V

$$VOUT = (VFB \cdot RFB1) / RFB2 + VFB = 4.99V$$

5V-->3.3V

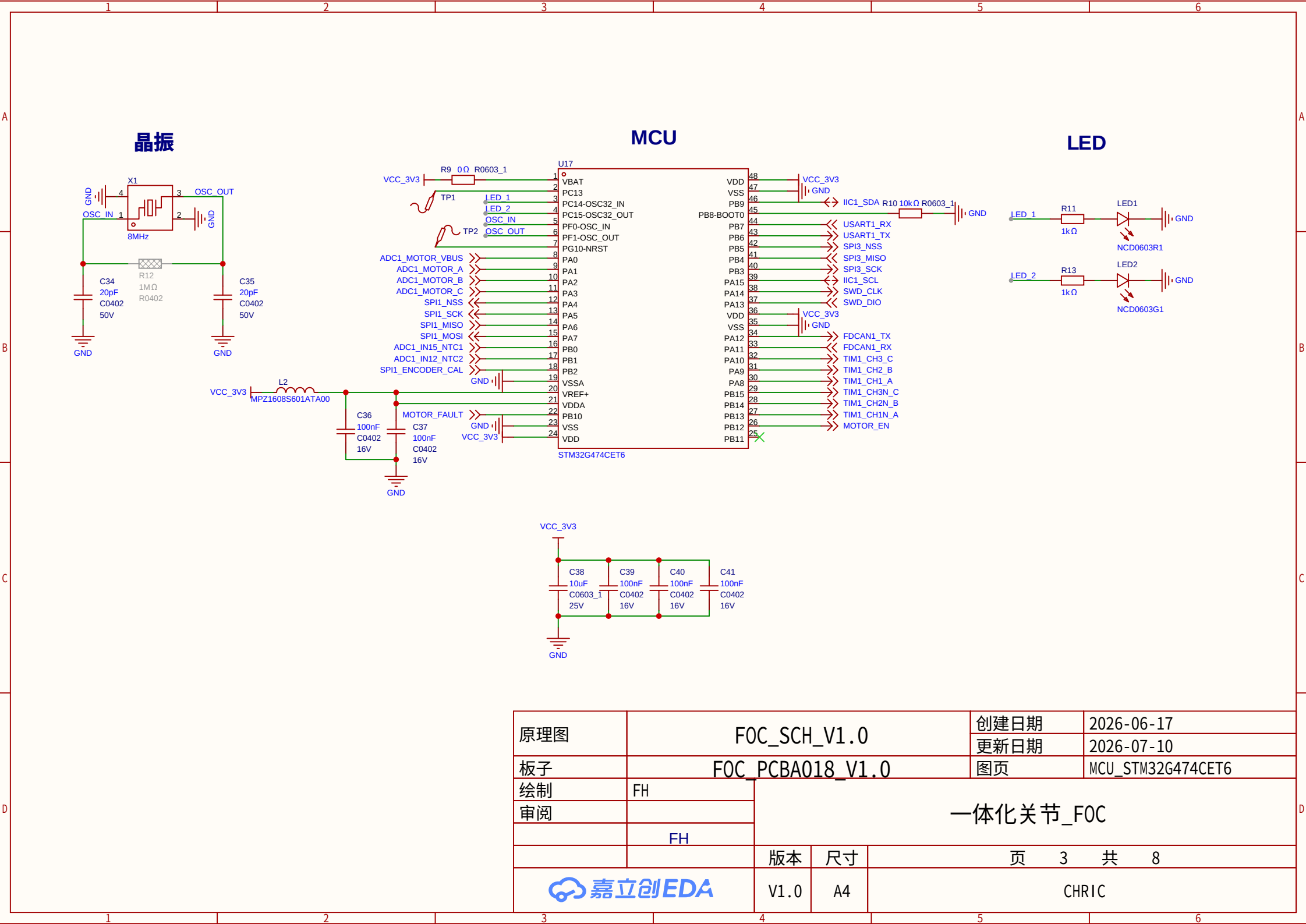


MCU供电

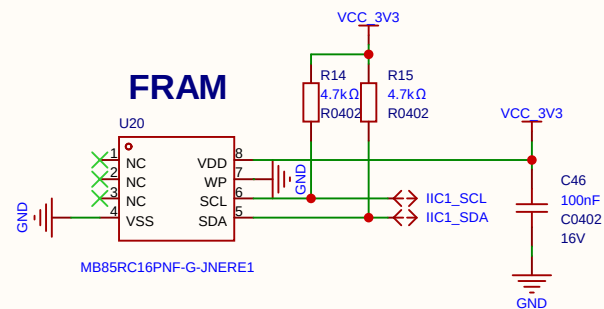
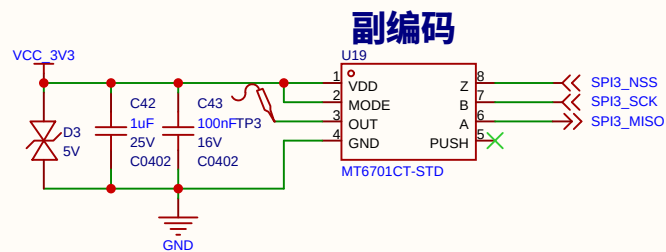
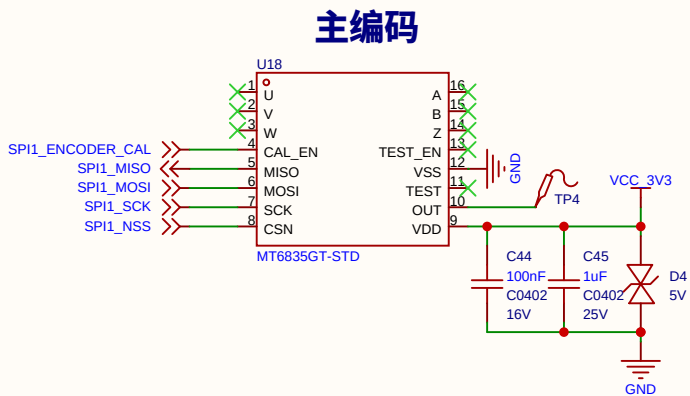


ADC基准供电

|        |                  |           |      |            |
|--------|------------------|-----------|------|------------|
| 原理图    | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-16 |
| 板子     | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-09 |
| 绘制     | FH               | 一体化关节_FOC |      |            |
| 审阅     |                  |           |      |            |
|        | FH               |           |      |            |
|        |                  | 版本        | 尺寸   | 页 2 共 8    |
| 嘉立创EDA |                  | V1.0      | A4   | CHRIC      |



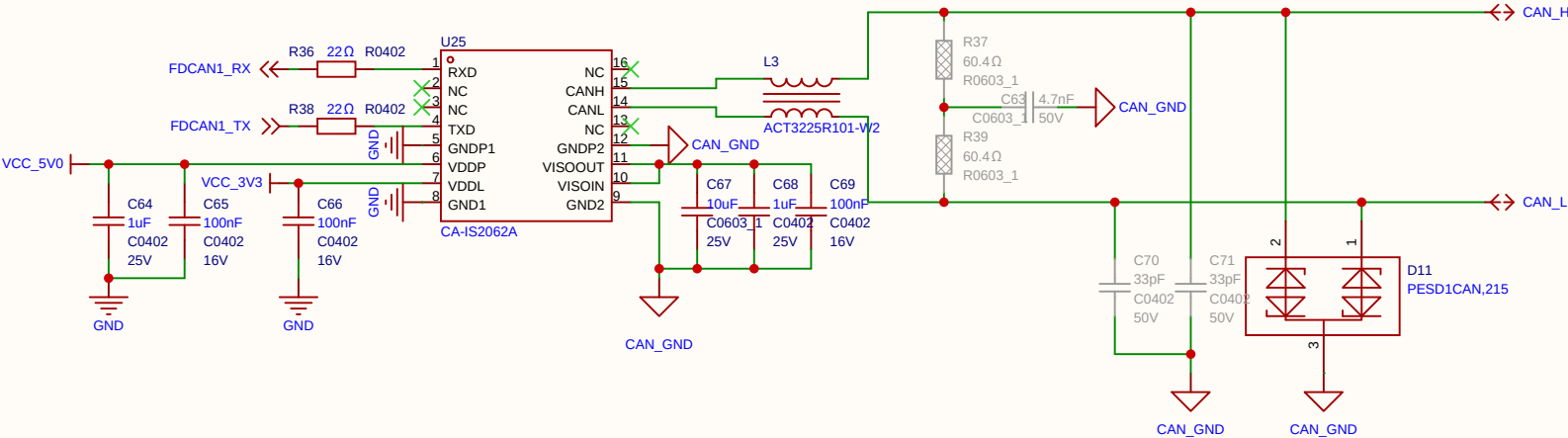
|        |                  |           |      |            |
|--------|------------------|-----------|------|------------|
| 原理图    | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-17 |
| 板子     | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-10 |
| 绘制     | FH               | 一体化关节_FOC |      |            |
| 审阅     |                  |           |      |            |
|        | FH               |           |      |            |
|        |                  | 版本        | 尺寸   | 页 3 共 8    |
| 嘉立创EDA |                  | V1.0      | A4   | CHRIC      |



|     |                  |           |      |            |
|-----|------------------|-----------|------|------------|
| 原理图 | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-12 |
| 板子  | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-08 |
| 绘制  | FH               | 一体化关节_FOC |      |            |
| 审阅  |                  |           |      |            |
|     | FH               |           |      |            |
|     |                  | 版本        | 尺寸   | 页 4 共 8    |
|     |                  | V1.0      | A4   | CHRIC      |

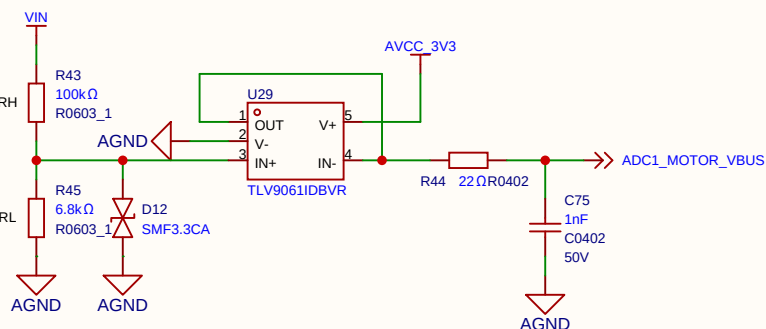
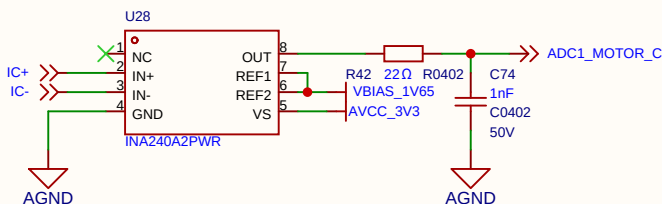
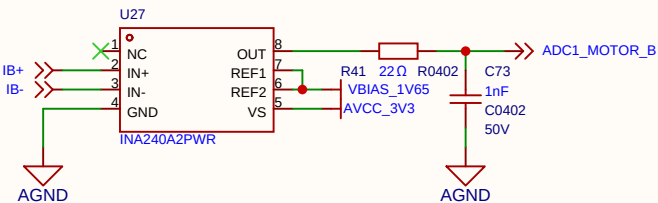
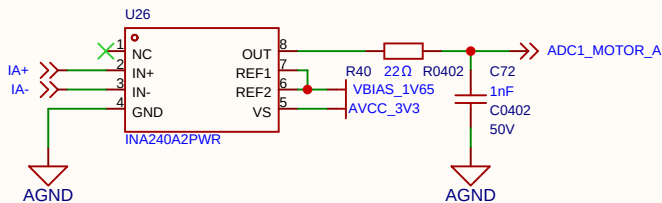


CAN隔离收发



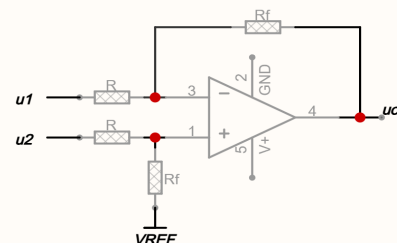
|        |                  |           |      |            |
|--------|------------------|-----------|------|------------|
| 原理图    | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-12 |
| 板子     | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-08 |
| 绘制     | FH               | 一体化关节_FOC |      |            |
| 审阅     |                  |           |      |            |
|        | FH               |           |      |            |
|        |                  | 版本        | 尺寸   | 页 6 共 8    |
| 嘉立创EDA |                  | V1.0      | A4   | CHRIC      |

## 差分放大



$$VBUS(VIN) = VBUS\_DET * (RH + RL) / RL$$

## 电流采样计算



$$\text{运放增益: } k = R_f / R$$

$$\text{偏置电压 (应对负电压): } V_{REF} = 1.65V = 2048$$

$$\text{采样电阻: } R_m = 1m\Omega$$

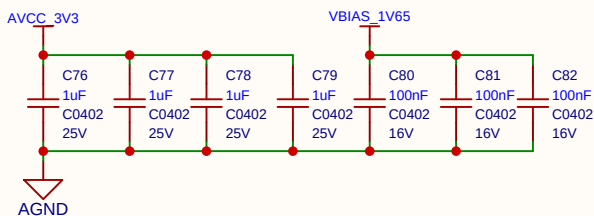
$$\text{运放输出: } u_o = k * (u_2 - u_1) + V_{REF}$$

$$u_2 - u_1 = I * R_m$$

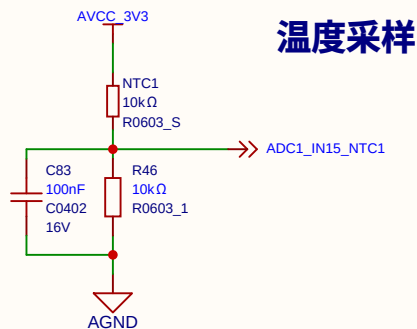
$$u_o = (adc\_val - 2048) * 3.3V / 4096$$

$$\text{采集电流: } I = u_o / (R_m * k)$$

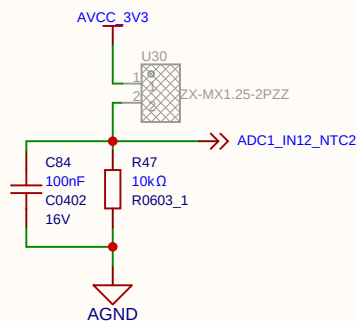
$$ADC \text{ } I_{max} \rightarrow 41.25A (3.3V)$$



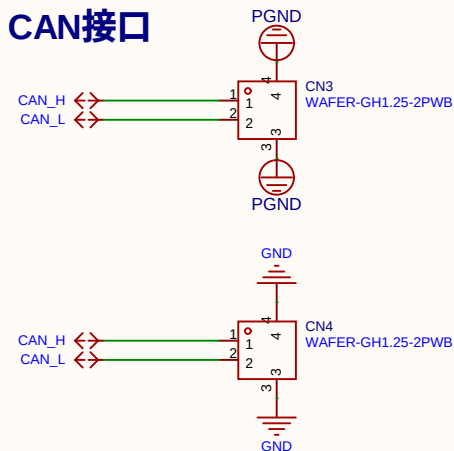
|     |                  |           |      |            |
|-----|------------------|-----------|------|------------|
| 原理图 | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-17 |
| 板子  | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-09 |
| 绘制  | FH               | 一体化关节_FOC |      |            |
| 审阅  |                  |           |      |            |
|     | FH               |           |      |            |
|     |                  | 版本        | 尺寸   | 页 7 共 8    |
|     |                  | V1.0      | A4   | CHRIC      |



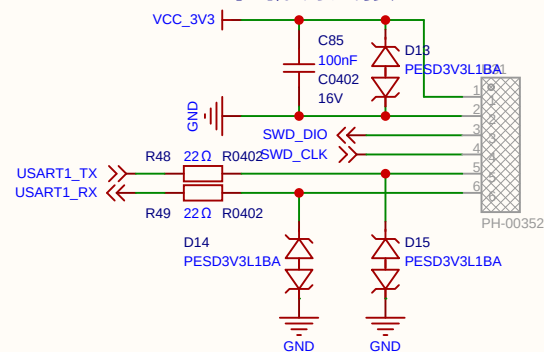
## 温度采样



## CAN接口



## 下载调试接口



|        |                  |           |      |            |
|--------|------------------|-----------|------|------------|
| 原理图    | FOC_SCH_V1.0     |           | 创建日期 | 2026-06-15 |
| 板子     | FOC_PCBA018_V1.0 |           | 更新日期 | 2026-07-08 |
| 绘制     | FH               | 一体化关节_FOC |      |            |
| 审阅     |                  |           |      |            |
|        | FH               |           |      |            |
|        |                  | 版本        | 尺寸   | 页 8 共 8    |
| 嘉立创EDA |                  | V1.0      | A4   | CHRIC      |